

Auteur: Stan Van Campenhout

Studierichting: Informatica

Oefeningenreeks 4A nr. 60.8

$$\begin{aligned} & \int \frac{\sqrt{x} - 1}{(\sqrt[3]{x} - 1)\sqrt[6]{x}} dx \stackrel{t=\sqrt[6]{x}}{=} \int \frac{t^3 - 1}{t(t^2 - 1)} \cdot 6t^5 dt \\ &= 6 \int \frac{t^8 - t^5}{t(t^2 - 1)} dt = 6 \int \left(t^5 + t^3 - t^2 + t - 1 + \frac{t^2 - t}{t^3 - t} \right) dt \\ &= t^6 + \frac{3}{2}t^4 - 2t^3 + 3t^2 - 6t + 6 \int \frac{t(t-1)}{t(t-1)(t+1)} dt \\ &= t^6 + \frac{3}{2}t^4 - 2t^3 + 3t^2 - 6t + 6 \ln |t+1| + c \\ &\stackrel{t=\sqrt[6]{x}}{=} x + \frac{3}{2}x^{\frac{2}{3}} - 2x^{\frac{1}{2}} + 3x^{\frac{1}{3}} - 6x^{\frac{1}{6}} + 6 \ln |x^{\frac{1}{6}} + 1| + c \end{aligned}$$

References